ANALYTICAL CUME
October 29th 2004

NASA's twin robot geologists, the Mars Exploration Rovers, were launched toward Mars on June 10 and July 7, 2003, in search of answers about the history of water on Mars. They landed on Mars on January 3 and January 24 PST. The Mars Exploration Rover mission is part of NASA's Mars Exploration Program, a long-term effort of robotic exploration of the red planet. The fall 2004 Barringer Lecturer was Daniel McCleese, Director of the NASA Mars Exploration Program.

Primary among the mission's scientific goals is to search for and characterize a wide range of rocks and soils that hold clues to past water activity on Mars. The spacecraft are targeted to sites on opposite sides of Mars that appear to have been affected by liquid water in the past. The landing sites are at Gusev Crater, a possible former lake in a giant impact crater, and Meridiani Planum, where mineral deposits (hematite) suggest Mars had a wet past.

The science instruments on the rovers are three very different instruments that demonstrate different applications of analytical spectroscopy. They are:

Miniature Thermal Emission Spectrometer (Mini-TES)
Mössbauer Spectrometer (MB)
Alpha Particle X-Ray Spectrometer (APXS)

Write an essay explaining the scientific principles behind each of these instruments and their application to the robotic exploration of Mars. Be quantitative in your explanation and explain the type of scientific data that can be obtained. Diagrams and examples will help your explanation. End you essay by explaining some of the major results of the mission to date.

Scoring:

Mini-TES explanation 20%.
MB explanation 20%.
APXS explanation 20%.
Use of quantitative examples 10%.
Description of scientific data that can be obtained 10%.
Quality of diagrams and examples 10%.
Summary of results to date 10%.

Pass mark: 80%